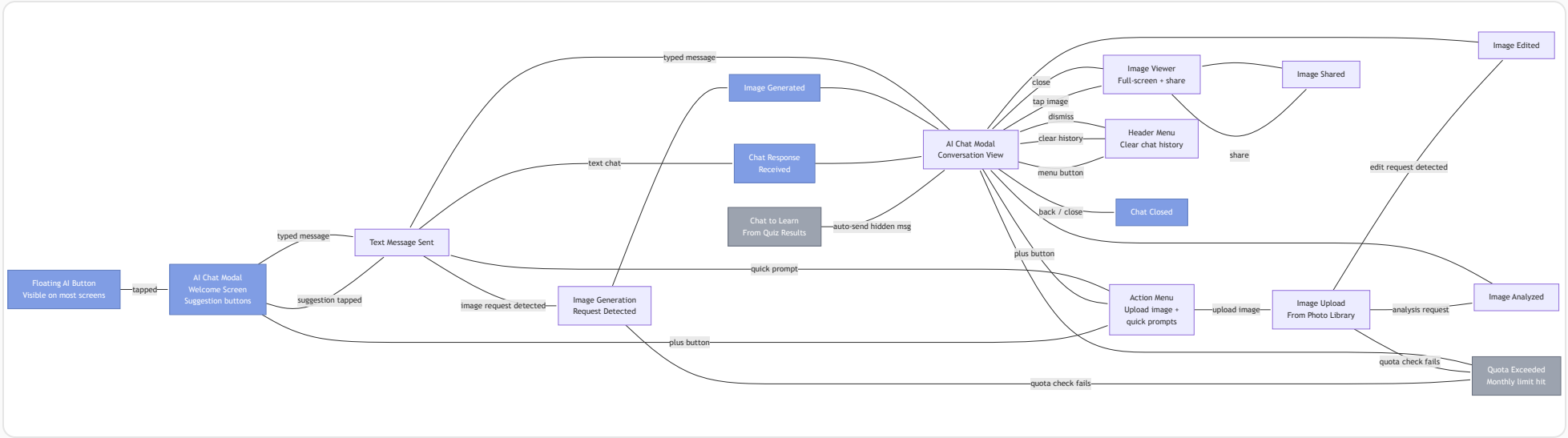


Archives — AI Chat Flow Spec

AFF-844 · PostHog Project 93650 · Generated 2026-04-30

- ☐ **Normal** — event exists in codebase
- ☐ **Red** — not in codebase, needs implementation
- ☐ **Grey** — in codebase but not wired
- ☒ **Blue** — checkpoint (key milestone)
- ☐ **Dark Grey** — alternate path

User Journey



Events

#	Node	PostHog Events	QA
1	Floating AI Button	ai_button_visible screen (current route pathname) ai_button_tapped screen (current route pathname)	Button not visible; button overlaps content; tap doesn't open chat; drag gesture blocks tap; button disappears after drag
2	Chat to Learn (from Quiz Results)	quiz_results_chat_to_learn_clicked adventure_id, module_id, previous_score, era_id, era_name chat_to_learn_paywall_shown adventure_id, module_id, era_id, era_name, trigger: chat_to_learn subscribe_purchase_completed trigger: chat_to_learn subscribe_purchase_cancelled trigger: chat_to_learn chat_to_learn_response era_id, response_length	Paywall not presented; hidden message not sent to AI; quota error before API call; response not displayed; quiz context banner missing
3	AI Chat Modal — Welcome Screen	ai_chat_opened era_id, message_count ai_suggestion_tapped suggestion_text, era_id	Welcome screen not shown when messages is empty; character image missing; suggestion buttons unresponsive; speech bubble text cut off
4	AI Chat Modal — Conversation View	ai_chat_opened era_id, message_count ai_chat_history_loaded message_count, source (cloud local migration)	Chat history not loading; messages duplicated after reload; scroll doesn't jump to latest message; markdown rendering broken

#	Node	PostHog Events	QA
5	Text Message Sent	ai_chat_message_sent era_id, message_type (text image), message_length, is_first_message	Send button disabled when it should be active; keyboard doesn't dismiss; empty message gets sent; input text not cleared after send
6	Chat Response Received	ai_chat_response_received era_id, response_length, has_web_sources, web_sources_count ai_chat_error era_id ai_web_source_tapped source_url, source_title, era_id	Response not displayed; loading spinner stuck; error message not shown; web sources links broken; markdown not rendering
7	Image Generation Request	ai_image_generated era_id ai_chat_error era_id ai_image_generation_requested era_id, prompt_length	Image request not detected by keyword matcher; "Generating image..." spinner stuck; generated image not displayed; base64 decode failure
8	Image Upload (Photo Library)	ai_image_selected era_id, source: library ai_image_upload_cancelled era_id	Permission denied on iOS; image picker doesn't launch; base64 encoding fails; pending image preview not shown; remove button doesn't clear
9	Image Analyzed	ai_image_analyzed era_id, has_question ai_chat_error era_id	"Analyzing image..." spinner stuck; analysis response empty; uploaded image not sent to API; error not surfaced to user

#	Node	PostHog Events	QA
10	Image Edited	ai_image_edited era_id ai_chat_error era_id	Edit request not detected by keyword matcher; "Generating image..." spinner stuck; edited image not displayed; upload to Supabase storage fails
11	Quota Exceeded	ai_quota_exceeded request_type (chat image_generate image_edit image_analyze), limit, is_subscriber	Quota error message not shown; wrong limit displayed; reset date incorrect; subscriber flag wrong (shows free-tier limit for paid user)
12	Action Menu	ai_action_menu_opened era_id ai_quick_prompt_tapped prompt_text, era_id	Menu doesn't appear; menu items unresponsive; cancel button doesn't dismiss; quick prompt sends wrong text
13	Image Viewer (Full-screen)	ai_image_shared era_id ai_image_viewer_opened era_id, image_type (generated edited uploaded)	Image viewer blank; full-screen image not loading from base64; share button doesn't open share sheet; temp file cleanup fails; close button unresponsive
14	Header Menu (Clear History)	ai_chat_history_cleared era_id, message_count (before clear)	Menu doesn't appear; clear confirmation alert doesn't show; messages not actually deleted from Supabase; menu doesn't dismiss after action
15	Chat Closed	ai_chat_closed era_id, message_count ai_chat_session_duration duration_seconds, messages_sent, era_id	Modal doesn't close; messages not saved to cloud before close; back button on Android doesn't trigger close tracking

AI Quiz Explanation events (related but outside chat modal)

Event	Properties	Status
ai_quiz_explanation_requested	adventure_id, module_id, era_name, question_count	Wired in AIQuizExplanation.tsx
ai_quiz_explanation_generated	adventure_id, module_id, era_name, question_count, duration_ms	Wired in AIQuizExplanation.tsx

Questions This Flow Answers

1. Strategic Questions

- 1. How many users open the AI chat at least once per week?
- 2. How many users open AI chat from the floating button?
- 3. How many users open AI chat from the "Chat to Learn" button on quiz results?
- 4. How many users hit the monthly quota limit?
- 5. How many free users hit the quota limit and then see the paywall?
- 6. How many users who hit the quota limit upgrade to a paid plan?
- 7. How many AI chat sessions result in zero messages sent (user opened and closed immediately)?
- 8. How many users send image-related requests (generation, editing, analysis)?
- 9. How many users use the AI chat more than 5 times in a single day?

2. Behavioral Insights

- 1. How many users tap a suggestion button on the welcome screen as their first interaction?
- 2. How many users type a custom first message instead of using suggestions?
- 3. How many users send a text message in the AI chat?

4. How many users receive a chat response with web search sources?
5. How many users tap on a web search source link?
6. How many users request image generation through text prompts?
7. How many users upload an image from their photo library?
8. How many uploaded images are analyzed (asked about) rather than edited?
9. How many uploaded images are edited (transformed)?
10. How many users open the full-screen image viewer?
11. How many users share a generated or edited image?
12. How many users open the action menu (plus button)?
13. How many users select a quick prompt from the action menu?
14. How many users clear their chat history?
15. How many users encounter an AI chat error?
16. How many users dismiss the Chat to Learn paywall without subscribing?
17. How many users complete a Chat to Learn session (quiz context + AI response)?
18. How many users send a follow-up message after a Chat to Learn response?

3. Checkpoint Validation

1. **Floating AI Button** — How many users see the floating button on app screens?
2. **AI Chat Modal (Welcome)** — How many users open the chat modal and see the welcome screen?
3. **Chat Response Received** — How many users send a message and get a successful AI response?
4. **Image Generated** — How many users request and receive a generated image?
5. **Chat Closed** — How many users close the chat after at least one exchange?

Analysis Notes

- **Chat to Learn is gated by subscription:** Non-subscribers see a paywall when tapping "Chat to Learn" on quiz results. Track conversion rate from `chat_to_learn_paywall_shown` to `subscribe_purchase_completed` with trigger `chat_to_learn`.
- **Quota enforcement moved server-side:** The client-side `checkQuota()` in `AIStorageService` now always returns `allowed: true`. Actual quota enforcement is handled by the backend API (returns HTTP 429). The `ai_quota_exceeded` event fires on client when the backend rejects a request.
- **Image request detection is heuristic:** The `isImageRequest()` and `isImageEditRequest()` methods use keyword matching. Some user prompts may be misclassified (image intent routed to text chat or vice versa). Adding `ai_image_generation_requested` would help measure false-negative rate.
- **Session-based storage:** Messages are stored per-session in Supabase `ai_user_data` table. The `ai_chat_session_duration` event would enable session-length analysis without relying on timestamp math between open/close events.
- **Floating button is hidden on certain screens:** The button is hidden on onboarding, auth, today tab, and subscribe screens. Track `ai_button_visible` to understand which screens drive the most chat opens.
- **Cloud sync for messages:** Messages sync to Supabase with 1.5s debounce. If the user closes the app immediately after sending, the last message may not sync. The `ai_chat_history_loaded` event with `source` would help diagnose sync issues.